Kinetic and potential energy scaling in vertically vibrated granular beds: ballistic–collisional subdomains and a smooth energy-partition crossover from data-driven dimensional analysis

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How the kinetic and potential energies of a vertically vibrated granular bed scale with vibration strength Γ, amplitude-to-diameter ratio a/d, and fill height F remains open: existing scaling studies address single responses over restricted ranges, and existing phase diagrams are morphological rather than predictive. We combine 516 discrete-element simulations spanning $\Gamma \in [1.25, 160]$, $a/d \in [0.25, 4]$, $F \in [4, 16]$ with data-driven dimensional analysis (smoothed partial derivatives, subtractive clustering, fuzzy c-means, per-subdomain power-law fitting). The dimensionless kinetic energy e_k^* admits a compact two-subdomain pure power law: $e_k^* = 2.958\,(a/d)^{0.564}\Gamma^{0.717}F^{-0.811}$ (collisional, 278 cases, $\delta = 18.0\%$) and $e_k^* = 2.059\,(a/d)^{1.044}\Gamma^{0.629}F^{-0.554}$ (ballistic, 106 cases, $\delta = 29.4\%$), with global error $\delta^* = 21.1\%$ versus 23.5% single-domain. The potential-energy proxy e_p^* (mean elevation in diameters) resists compact partitioning: six subdomains with no stable cluster-count plateau, exponents on a/d flipping from -0.22 to $+0.78$ and on F from -1.05 to $+0.58$ between neighbors ($\delta^* = 26.9\%$ versus 35.9% single-domain). Snapshot and time-averaged field diagnostics at $F = 8$ show the two kinetic subdomains differ in energy-injection mechanism—ballistic flight at large a/d or extreme Γ (coherent plugging bed or Fermi-accelerated Leidenfrost state with a levitated dense layer above a bottom gap), collisional boiling otherwise—cutting across morphological state categories. The partition ratio $R = e_p^*/(e_k^* + e_p^*)$ varies smoothly along the model line from 0.13 through a 0.60 plateau to 0.90 with increasing Γ; hold-out measurements at $\Gamma = 4$ and 15 straddle the plateau with no discontinuity. Six hold-out simulations validate the kinetic scaling with 16.5% mean absolute error. Mechanism-resolved subdomains thus replace morphological state labels with quantitative, transferable scaling criteria.

I. INTRODUCTION

Vertically vibrated granular beds display a rich spectrum of states—solid-like compaction, bouncing beds, undulations, buoyancy-driven convection, the granular Leidenfrost state, and granular gases [1–5]. Two complementary descriptions exist. The first is *morphological*: Eshuis *et al.* mapped phase boundaries of a shallow quasi-two-dimensional bed in (F, Γ) and (F, S) planes ($S = a^2\omega^2/gd$), identifying detachment, undulation, Leidenfrost, and convection thresholds from direct observation [2]. The second is *quantitative scaling*: granular temperature and mean kinetic energy have been reported to scale as powers of peak vibration velocity or Γ [6–8], most often for a single response variable over a restricted parameter range.

Missing is a joint quantitative account of how the two mechanical-energy reservoirs—kinetic and potential—scale across the full parameter space $(\Gamma, a/d, F)$, and whether scaling-regime boundaries coincide with morphological phase boundaries. The question matters beyond taxonomy: industrial vibrated-bed processes are designed by adjusting Γ, a, and fill level, and design rules based on mechanism-resolved energy scaling are more transferable than state labels.

Three unknowns define the gap. First, no prior study jointly fits the dimensionless mean kinetic energy e_k^* and a potential-energy proxy e_p^* (mean particle elevation in particle diameters) over a common parameter space; published scaling laws treat one response at a time [6, 7], and the partition between kinetic and potential reservoirs has been described only qualitatively [2]. Second, existing phase diagrams report observation-based boundaries [2, 4]; whether scaling exponents change abruptly at those boundaries—the signature of a mechanism switch—or vary smoothly is unknown. Third, a/d is known to matter independently of Γ [1, 8], but its interaction with F in multilayer beds is unresolved quantitatively.

We address these unknowns with data-driven dimensional analysis (DDDA): smoothed partial derivatives identify local power-law sensitivities, subtractive clustering with fuzzy c-means partitions the exponent field into subdomains, and each subdomain receives a closed-form power law [9, 10]. The approach is agnostic to preconceived phase labels and lets the data reveal how many regimes are actually needed. We ran 516 discrete-element simulations covering two decades of Γ, four octaves of a/d, and three fill heights, in an adaptive two-round design with boundary refinement and independent hold-out validation.

This paper contributes: (i) a two-subdomain pure power law for e_k^* whose subdomains differ in exponents on all three parameters; (ii) the demonstration that e_p^* resists compact partitioning—a continuously varying response with sign-flipping sensitivities; (iii) mechanistic identification of the kinetic subdomains as *ballistic* versus *collisional* energy injection, established by snapshots and time-averaged solid-fraction fields at $F = 8$, cutting across morphological categories; and (iv) a smooth

energy-partition landscape $R(\Gamma, a/d, F)$ with no internal discontinuity.

II. SYSTEM AND METHODS

A. Simulated system

We simulate a quasi-two-dimensional cell (length 0.10 m, height 0.20 m, width 4.5 mm) with monodisperse spheres ($d = 2$ mm, density 2500 kg/m^3) in the LIGGGHTS discrete-element platform. The base executes sinusoidal vertical motion $z_b(t) = a \sin \omega t$. The three dimensionless control parameters are vibration strength $\Gamma = a\omega^2/g$, amplitude-to-diameter ratio a/d, and nominal fill height F (rest layers; realized counts 106/231/502 particles for $F = 4/8/16$ after gravitational settling). Each run discards 20 development periods and averages over 8 statistical periods.

B. Response variables

The primary responses are

$$e_k^* = \frac{\langle \frac{1}{2}|\mathbf{u}|^2 \rangle}{gd}, \qquad e_p^* = \frac{\langle y \rangle}{d}, \tag{1}$$

the dimensionless mean kinetic energy and the mean particle elevation above the base in units of d. Because e_p^* is an elevation rather than a gravitational energy, all partition statements below are statements about the split between elevation and motion on the common scale gd. The vertical granular temperature t_y^* and center-of-mass height h_m^* are analyzed for comparison. The partition ratio is $R = e_p^*/(e_k^* + e_p^*)$.

C. Design of simulations

Round 1: full factorial grid, Γ at 15 log-spaced levels in $[1.25, 160]$, a/d at 9 log-spaced levels in $[0.25, 4]$, $F \in \{4, 8, 16\}$—405 cases. Round 2: boundary refinement (5 new a/d levels $\times$ 7 new Γ levels $\times$ 3 F, 105 cases) plus 6 hold-out validation points at $(a/d, \Gamma) = (0.45, 4), (0.45, 15), (0.85, 4), (0.85, 15), (1.5, 4), (1.5, 15)$, $F = 8$. Merging all runs and de-duplicating repeated parameter combinations (105 duplicates) leaves 516 unique valid cases: 405 from round 1 and 111 from round 2 (105 refinement plus 6 hold-out, one of which duplicated a refinement point). No case failed.

D. Data-driven dimensional analysis

DDDA proceeds in three steps. (1) Smoothed partial derivatives $\partial \log Y/\partial \log X_i$ are estimated by adaptive Gaussian-weighted local linear regression in log space. (2) Subtractive clustering (radius factor r_a) plus fuzzy c-means ($m = 2$) partitions the exponent field; points with maximum membership below 0.6 are ambiguous. (3) Each cluster defines a subdomain; within it the composite $C = (a/d)^{D_1}\Gamma^{D_2}F^{D_3}$ follows from the dominant eigenvector, and the physical-space law $Y = A\,C^b$ (pure power law, $c = 0$) is fitted with iterative residual-based outlier cleaning. The global error metric is

$$\delta^* = \frac{1}{N}\sum_i \frac{|Y_i^{\mathrm{pred}} - Y_i|}{Y_i}. \tag{2}$$

The radius r_a controls resolution; we sweep $r_a \in \{0.2, 0.3, 0.4, 0.5\}$ and require stable cluster counts across neighbors before accepting a partition. Single-cluster fits ($n = 1$) provide baselines. Subdomain membership at new points is predicted by 7-nearest-neighbor voting in log space, which also yields interfacial vote fractions.

III. RESULTS AND DISCUSSION

A. Kinetic energy: a robust two-subdomain power law

For e_k^* the DDDA partition is stable: at $r_a = 0.3$ the merged 516-case dataset resolves two subdomains with global error $\delta^* = 21.1\%$, versus 23.5% for a single-domain fit ($e_k^* = 2.540\,(a/d)^{0.647}\Gamma^{0.698}F^{-0.700}$, $R^2 = 0.956$)—a 10% relative improvement [Fig. 1(a)]. The subdomain laws are

$$\text{subdomain 0 (collisional):}\ \ e_k^* = 2.958\,(a/d)^{0.564}\Gamma^{0.717}F^{-0.811}, \tag{3}$$

$$\text{subdomain 1 (ballistic):}\ \ e_k^* = 2.059\,(a/d)^{1.044}\Gamma^{0.629}F^{-0.554}. \tag{4}$$

Subdomain 0 carries 308 of 516 cases (59.7%), of which 278 survive outlier cleaning, with $\delta = 18.0\%$, $R^2 = 0.982$; subdomain 1 carries 117 cases (22.7%), 106 after cleaning, with $\delta = 29.4\%$, $R^2 = 0.810$. The remaining 91 cases (17.6%) fall below the membership threshold, concentrated in the transition band. The decisive evidence for two subdomains is not the modest error reduction but the *exponent shift*: the a/d exponent nearly doubles ($0.564 \to 1.044$) while the Γ and F exponents change little.

Three checks support Eqs. (3)–(4). First, the round-1 fits (405 cases, subdomains of 225 and 109 cases) reproduce the merged-data exponents closely: the subdomain-0 exponents shift by at most 0.03 (a/d: $0.547 \to 0.564$), while the subdomain-1 a/d exponent shifts by 0.095 ($0.949 \to 1.044$) with the Γ and F exponents stable to 0.07 ($0.705 \to 0.629$, $0.624 \to 0.554$)—demonstrating insensitivity to boundary refinement at the level of the exponent precision. Second, resolution sweeps show the cluster count is stable at 2 for $r_a = 0.3$–0.4 while finer radii ($r_a \le 0.2$) over-segment. Third, the six hold-out

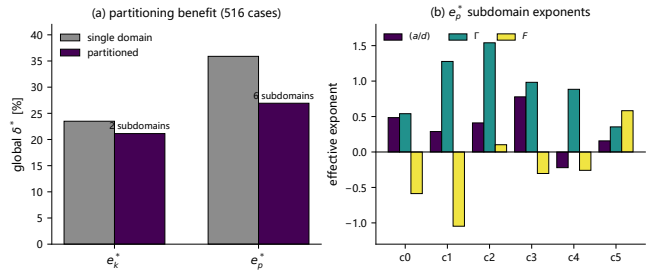


FIG. 1. (a) Global relative error δ^* of single-domain versus partitioned power-law fits on the merged 516-case dataset: e_k^* needs only two subdomains; e_p^* fragments into six with no stable plateau. (b) Effective exponents of the six e_p^* subdomains, showing sign flips on a/d (c4: -0.221) and F (c2: $+0.103$, c5: $+0.582$, versus c1: -1.047).

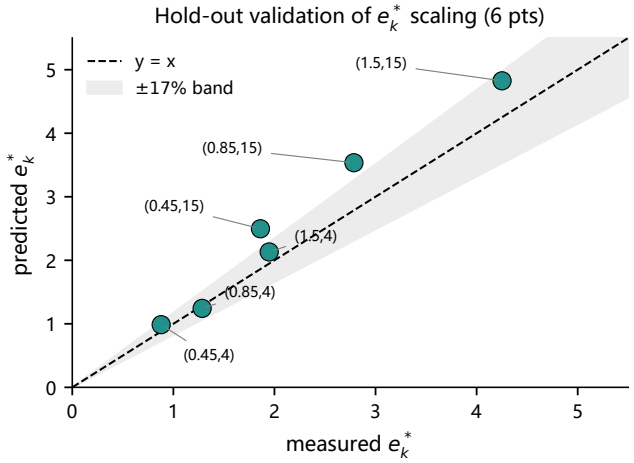


FIG. 2. Hold-out validation of the e_k^* scaling. Six independent simulations (labels give $(a/d, \Gamma)$, $F = 8$) against predictions from the round-1 model. The shaded band marks ±17%, the round-1 subdomain error scale; the mean absolute error is 16.5%.

simulations—run after the model was frozen, at $(a/d, \Gamma)$ combinations absent from the training grid—validate the scaling: predictions lie within -3.2% to $+34.1\%$ of the measurements, with mean absolute error 16.5% and mean signed error $+15.5\%$ (Fig. 2). The systematic overprediction concentrates at $\Gamma = 15$ inside the transition band, consistent with the subdomain interface running between the $\Gamma = 4$ and $\Gamma = 15$ validation columns.

B. Potential energy: a response that resists partitioning

The potential-energy proxy e_p^* behaves qualitatively differently. At every resolution the optimal partition splinters: $r_a = 0.3$ gives 11 clusters ($\delta^* = 24.5\%$, 48% ambiguous), $r_a = 0.4$ gives 6 clusters ($\delta^* = 26.9\%$, 35.5% ambiguous), $r_a = 0.5$ gives 5 clusters ($\delta^* = 21.2\%$, 28.6% ambiguous, on the 405-case round-1 data). No stable plateau exists in the cluster count, and the six-cluster solution orders exponents non-monotonically across neighboring clusters [Fig. 1(b)]: the a/d exponent ranges from -0.221 (c4) to $+0.778$ (c3), the F exponent from -1.047 (c1) to $+0.582$ (c5). The single-domain baseline is poor ($\delta^* = 35.9\%$, $R^2 = 0.829$), so partitioning helps by 25% relative, but no compact law emerges.

The physical reading is that e_p^* varies by smooth crossover rather than by switching between a few mechanisms. The smoothed derivative field makes this explicit: the local a/d sensitivity spans $[-0.675, +1.534]$ and the local F sensitivity spans $[-1.432, +0.812]$, with sign changes along continuous fronts rather than at isolated boundaries. Increasing a/d raises the bed at moderate Γ but lowers the mean elevation in the deep-bed low-Γ corner (cluster 4's negative a/d exponent: thick beds loft less per unit drive); increasing F raises the mean elevation at low Γ (cluster 5's positive F exponent) but lowers it in the strongly agitated regime (cluster 1: energy diluted among more particles). These opposing trends superpose and interpolate, defeating low-dimensional partitioning.

The contrast between e_k^* (two clean subdomains) and e_p^* (no compact partition) is itself a result: the two reservoirs sample the mechanism landscape differently. Kinetic energy is set by the injection mechanism at the bed base, which switches sharply between ballistic and collisional regimes; potential energy integrates the entire bed-height distribution, which deforms continuously as the bed fluidizes and expands.

C. Subdomain geometry and the failure of single-parameter criteria

Where does the e_k^* boundary run? Probing the trained classifier on a $(a/d, \Gamma)$ grid at $F = 8$ (Fig. 3) shows subdomain 1 (ballistic) occupying two opposite corners: large a/d at low Γ ($a/d \gtrsim 0.6$, $\Gamma \lesssim 5$) and small a/d at extreme Γ ($a/d \lesssim 0.3$, $\Gamma \gtrsim 80$). Subdomain 0 (collisional) fills the diagonal band between, including the whole $a/d = 0.25$ column up to $\Gamma \approx 40$ and the whole $\Gamma = 20$ row. Along F, the boundary moves monotonically: at fixed $(a/d, \Gamma) = (0.25, 1.25)$, $F = 4$ belongs to subdomain 0 while $F = 16$ belongs to subdomain 1, and likewise at $(1.0, 2.5)$ and $(2.0, 5.0)$—deep beds stay solid-like to higher Γ, consistent in trend with the Rayleigh–Janssen detachment criterion $\Gamma_{\mathrm{detach}} = 2 - e^{-\chi}$, whose critical value rises with bed depth [2].

Two features deserve emphasis. First, the interface is *diffuse*: at boundary points the 7-nearest-neighbor votes split (4:3 at $a/d = 1$, $\Gamma = 5$ and 80), and 17.6% of cases fall below the membership threshold. Second, the boundary is *not* a single-parameter criterion: constant-Γ lines fail (the boundary spans $\Gamma \approx 2.5$ to beyond 160), and constant-$\Gamma(a/d)^p$ lines also fail, as shown by the negative-slope guides in Fig. 3, which run opposite to the corner-

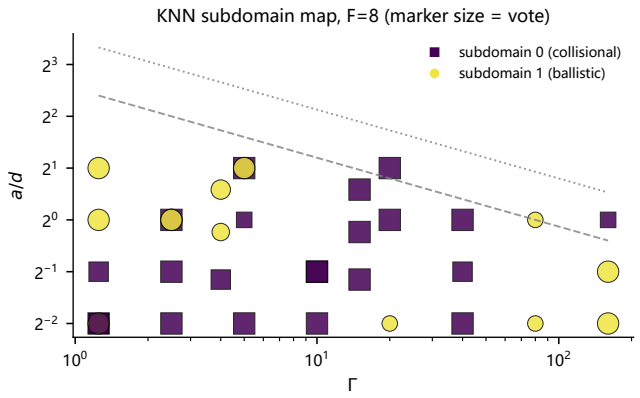


FIG. 3. Subdomain membership map for e_k^* in the $(\Gamma, a/d)$ plane at $F = 8$, probed by 7-neighbor voting on the trained DDDA classifier. Squares: subdomain 0 (collisional); circles: subdomain 1 (ballistic); marker size encodes the majority vote fraction (small markers = interfacial). Gray dashed/dotted lines: constant-$\Gamma(a/d)^{2.5}$ guides, drawn to show that no single power-law criterion matches the corner-shaped boundary.

shaped interface. The two-parameter structure reflects the two distinct routes into the ballistic regime documented next.

D. Mechanism: ballistic versus collisional injection at $F = 8$

Snapshot and field diagnostics at $F = 8$ (231 particles) resolve what the subdomains mean physically (Fig. 4). The subdomain-1 case at $(a/d, \Gamma) = (1, 2.5)$ is a solid-like plug: a dense bed with flat surface (top at 12–15 mm), particle velocities uniform at 0.1–0.3 m/s—of the order of the base velocity scale $\sqrt{\Gamma g a} \approx 0.22$ m/s—with a few fast particles at the base absorbing the landing impact. The bed takes off and lands as a body; its kinetic energy is *coherent translational* energy set by the base velocity. The subdomain-0 case at $(1, 40)$ is a collisional boiling bed: a dense phase topped by a dilute spray, collision hot spots (0.7–1.1 m/s) at the base, a few particles launched 2–3 diameters above the surface; energy dissipates through inelastic collisions within the bed.

The other route into subdomain 1, at $(0.25, 160)$, looks morphologically completely different—a dispersed cloud filling about three quarters of the cell—yet the time-averaged solid-fraction field (Fig. 5) shows a levitated layer (peak horizontally averaged fraction 0.09 at $y = 18$ mm) above a 12-mm bottom gap, with plume-like streaks rising to a cloud top of 116 mm: a granular Leidenfrost state [3, 4]. Particle speeds in the snapshot span 0.1–1.0 m/s, with the fastest particles in the dilute top spray, consistent with launch speeds of order the base velocity scale $\omega a \approx 0.89$ m/s—a Fermi-like acceleration picture [2, 3]. For comparison, the $F = 4$ case at the same $(a/d, \Gamma)$ shows the same structure with a slightly thicker

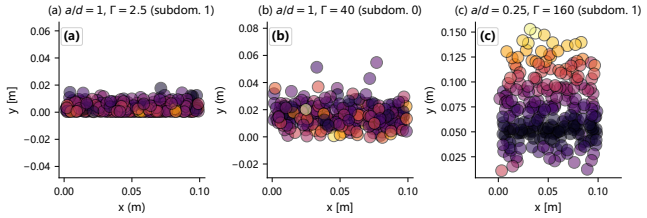


FIG. 4. End-frame snapshots colored by particle speed for three representative $F = 8$ cases (231 particles each): (a) $(a/d, \Gamma) = (1, 2.5)$, subdomain 1—coherent plugging bed (ballistic, solid-like); (b) $(1, 40)$, subdomain 0—collisional boiling bed with dilute spray; (c) $(0.25, 160)$, subdomain 1—dispersed granular gas / Leidenfrost state (ballistic, gas-like). Panels (a) and (c) share a scaling subdomain despite opposite morphology.

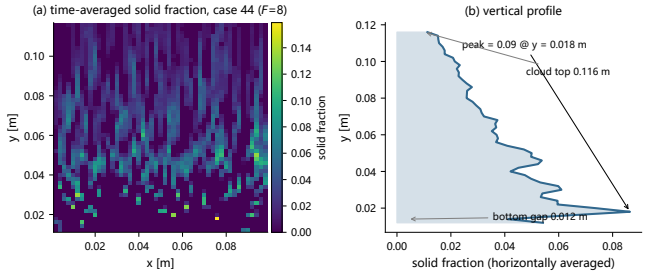


FIG. 5. Time-averaged solid-fraction field (a) and horizontally averaged vertical profile (b) for the Leidenfrost case $(a/d, \Gamma) = (0.25, 160)$, $F = 8$. A 0.012 m gap separates the base from a levitated layer peaking at 0.09 near $y = 0.018$ m; above a shoulder near $y = 0.04$–0.05 m the profile decays monotonically to a cloud top at 0.116 m.

gap (∼20 mm) and higher levitated peak (fraction 0.08 at $y = 25$ mm), so the Leidenfrost morphology is robust across fill heights in this corner of parameter space.

The mechanistic unification rests on the a/d exponent. In a ballistic regime the particle velocity inherits the base velocity scale $v \sim \omega a = \sqrt{\Gamma g a}$, so $e_k^* \propto (\omega a)^2/(gd) \propto \Gamma\,(a/d)$ at fixed F: unit a/d and unit Γ exponents before dilution among layers. Subdomain 1 matches the unit a/d exponent closely (1.044), while its Γ exponent (0.629) falls well below unity—consistent with partial dissipation of the injected energy, and with the F dilution factor ($F^{-0.554}$) absorbing part of the depth dependence; the ballistic identification therefore rests on the a/d exponent and the direct velocity evidence, not on a full unit-exponent match. In a collisional regime the bed stays dense, collisions randomize the energy, and the drive enters through a dissipation balance, softening the a/d exponent to 0.564. The subdomains thus classify *energy-injection mechanisms*, not morphological states: the plugging bed [Fig. 4(a)] and the Leidenfrost cloud [Fig. 4(c)] sit in one subdomain because both feed energy ballistically from the base, while the boiling bed [Fig. 4(b)] thermalizes it collisionally. This explains why

the e_k^* boundary cuts across the morphological phase diagram of Eshuis *et al.* [2] rather than retracing it, and why single-parameter criteria fail.

As a caveat, the exponent identification rests on fits with $\delta \sim 18$–29%; the ballistic interpretation of subdomain 1 is favored by the snapshot and field evidence and by the near-unit a/d exponent, but a dedicated study varying material dissipation would be needed to close the mechanism quantitatively.

E. Energy partition: a smooth crossover landscape

Combining both response models yields the partition ratio $R = e_p^*/(e_k^* + e_p^*)$ (Fig. 6). Along the model line $a/d = 0.5$, $F = 8$, R rises from 0.133 at $\Gamma = 2.5$ (kinetic-dominated coherent bouncing) through a plateau of ≈ 0.60 at $\Gamma = 10$–40 (collisional regime, the two reservoirs sharing the injected energy roughly equally) to 0.896 at $\Gamma = 160$ (potential-dominated Leidenfrost state, the bed lofted high).

The hold-out measurements probe the partition at two Γ values, 4 and 15, for three a/d values each. Pairwise at fixed a/d, R *decreases* from $\Gamma = 4$ to $\Gamma = 15$ (from 0.707 to 0.645 at $a/d = 0.45$; from 0.669 to 0.616 at 0.85; from 0.635 to 0.578 at 1.5)—opposite in sign to the model line's shallow rise (< 0.05 in R) over the same interval. The measured values (0.578–0.707) straddle the model plateau (≈ 0.60): the two Γ columns bracket the plateau from both sides, and the measured dip (0.05–0.06 in R) is comparable to the scatter expected from the subdomain fit errors ($\delta \sim 18$–29%). We therefore do not claim a confirmed monotone Γ trend from the hold-out data; what they support is (i) the absence of any discontinuity in R across the transition band, and (ii) the level of the plateau. Within each Γ column, smaller a/d gives larger R, consistent with the model's depth trend (deeper relative beds store more of their energy in elevation).

The partition landscape contains no jump. R varies smoothly across the parameter space, fully consistent with the non-partitionability of e_p^* (Sec. III B): both reservoirs deform continuously, and only the kinetic reservoir carries a signature of the ballistic–collisional switch. For applications this means the split of drive energy between agitation and lofting is a smoothly varying function of $(\Gamma, a/d, F)$ in this range, while the *character* of the agitation (coherent versus thermal) switches more sharply along the subdomain interface.

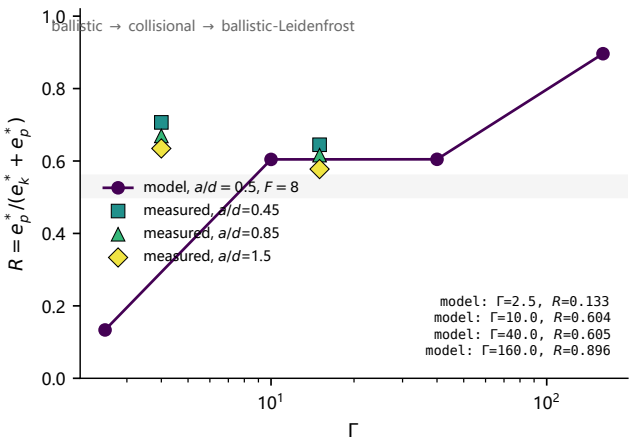


FIG. 6. Energy-partition ratio $R = e_p^*/(e_k^* + e_p^*)$ versus Γ. Line with circles: model prediction at $a/d = 0.5$, $F = 8$ (from the subdomain power laws); symbols: hold-out measurements at $a/d = 0.45$, 0.85, 1.5, $F = 8$. The model line crosses from kinetic-dominated ($R \approx 0.13$) through a ≈ 0.6 plateau to potential-dominated ($R \approx 0.9$); the hold-out pairs at $\Gamma = 4$ and 15 straddle the plateau level with no discontinuity.

IV. CONCLUSIONS

We mapped the joint kinetic–potential energy scaling of a vertically vibrated granular bed over $\Gamma \in [1.25, 160]$, $a/d \in [0.25, 4]$, $F \in [4, 16]$ using 516 discrete-element simulations and data-driven dimensional analysis. The kinetic energy e_k^* follows a two-subdomain pure power law [Eqs. (3)–(4)], validated on six hold-out cases to 16.5% mean absolute error; the subdomains correspond to ballistic and collisional energy injection—identified through the near-unit a/d exponent (1.044) plus direct velocity evidence—a mechanism-based boundary cutting across morphological phase categories and defeating single-parameter criteria. The potential-energy proxy e_p^* resists compact partitioning, with a/d and F sensitivities that flip sign along continuous fronts; the energy partition R consequently varies smoothly from 0.13 to 0.90 along the model line with no discontinuity, the hold-out pairs bracketing the mid-range plateau. These results replace morphological state labels with a two-layer description: a sharp switch in injection mechanism (kinetic energy) embedded in a continuous crossover landscape (energy partition).

Limitations bound the claims. All results hold for one material parameter set (fixed restitution and friction) and one quasi-two-dimensional geometry; material-parameter sweeps were outside our tool's reach and remain open. The potential-energy proxy is a mean elevation, not the full gravitational-energy distribution. Transition-band statistics rest on 8-period averaging, and interfacial points show vote splitting, so boundary locations carry $\mathcal{O}(10\%)$ uncertainty. The mechanism interpretation of the ballistic subdomain rests on the a/d exponent and snapshot/field evidence at three representative cases; direct dissipation-budget measurements are needed to close it quantitatively. Future work should test whether the ballistic a/d exponent stays near unity under restitution changes, and whether the R plateau at ≈ 0.6 is universal or material-dependent.

[1] H. M. Jaeger, S. R. Nagel, and R. P. Behringer, Granular solids, liquids, and gases, Rev. Mod. Phys. **68**, 1259 (1996).

[2] P. Eshuis, K. van der Weele, D. van der Hoef, M. Alder, and D. Lohse, Phase diagram of vertically shaken granular matter, Phys. Fluids **19**, 123301 (2007).

[3] B. Meerson, T. Pöschel, and Y. Bronfman, The granular Leidenfrost effect: experiment and theory of floating particle clusters, Phys. Rev. Lett. **95**, 258001 (2005).

[4] N. Rivas *et al.*, From the granular Leidenfrost state to buoyancy-driven convection, Phys. Rev. E **91**, 042202 (2015).

[5] P. Eshuis, K. van der Weele, D. van der Hoef, and D. Lohse, Onset of convection in strongly shaken granular matter, Phys. Rev. Lett. **104**, 038001 (2010).

[6] V. Zivkovic, M. J. Biggs, and D. H. Glass, Scaling of granular temperature in a vibrated granular bed, Phys. Rev. E **83**, 031308 (2011).

[7] S. S. Hsiau, L. S. Lu, and C. H. Tai, Experimental investigations of granular temperature in vertical vibrated beds, Powder Technol. **182**, 132 (2008).

[8] A. Bhateja and D. V. Khakhar, Vibrated granular bed: scaling of mean granular temperature, Phys. Rev. E **94**, 042901 (2016).

[9] P. G. Constantine, E. Dow, and Q. Wang, Active subspace methods in theory and practice: applications to kriging surfaces, SIAM J. Sci. Comput. **36**, A1500 (2014).

[10] E. Buckingham, On physically similar systems; illustrations of the use of dimensional equations, Phys. Rev. **4**, 345 (1914).